

Formal Analysis of the Erdős Unit Distance Problem

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Abstract

This document presents a rigorous mathematical breakdown of the Erdős Unit Distance Problem, which asks for the maximum number of unit distances that can exist among n points in the Euclidean plane. We employ Fourier analytic methods and incidence geometry bounding techniques.

Contents

1	Introduction and Contextual Literature Research	2
2	Analysis and Decomposition: Axiomatic Definitions	2
3	Autoformalization Architecture (Lean 4)	2
4	Fourier Analytic Derivation and Zero-Ellipse Lemmas	2
4.1	Empirical Measure and the Fourier Transform	2
4.2	Isolating the Principal Term	3
4.3	Bounding the Oscillatory Integral	3
4.4	Combinatorial Incidence Bound	4

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1 Introduction and Contextual Literature Research

The Erdős unit distance problem asks for the maximum possible number $u(n)$ of pairs of points in a set of n points in the Euclidean plane that are separated by a distance of exactly 1.

We frame our analysis using Fourier analytic methods, drawing parallels to the approaches pioneered by Iosevich and Rudnev for distinct distances on a sphere. By examining the Fourier transform of the empirical measure associated with the point set, we isolate the oscillatory constraints on repeated distances.

2 Analysis and Decomposition: Axiomatic Definitions

Definition 1 (Unit Distance Graph). *Let $P \subset \mathbb{R}^2$ with $|P| = n$. The unit distance graph $G(P) = (V, E)$ is defined by $V = P$ and $E = \{\{p, q\} \in P \times P \mid \|p - q\|_2 = 1\}$.*

The rigorous objective is to estimate the maximal cardinality of E . While known bounds rely on the Szemerédi-Trotter incidence theorem, Fourier analysis provides a deeper spectral description of distance densities.

3 Autoformalization Architecture (Lean 4)

```
import Mathlib.Data.Real.Basic
import Mathlib.Data.Finset.Basic
import Mathlib.Analysis.InnerProductSpace.EuclideanDist

open Finset
open EuclideanGeometry

def is_unit_distance (p q : EuclideanSpace \mathbb{R} (Fin 2)) : Prop
:=
  dist p q = 1

def unit_distances (P : Finset (EuclideanSpace \mathbb{R} (Fin 2))) :
  Finset (EuclideanSpace \mathbb{R} (Fin 2) \times EuclideanSpace \mathbb{R} (Fin 2)) :=
  P.product P |>.filter (fun x => x.1 \neq x.2 \land is_unit_distance x.1 x.2)

theorem erdos_unit_distance_bound (P : Finset (EuclideanSpace \mathbb{R} (Fin 2))) :
  \exists C : \mathbb{R}, C > 0 \land (unit_distances P).card \le C
    * (P.card : \mathbb{R}) ^ (4/3) := by
  sorry
```

4 Fourier Analytic Derivation and Zero-Ellipse Lemmas

4.1 Empirical Measure and the Fourier Transform

Let $\mu = \sum_{p \in P} \delta_p$ be the empirical Dirac measure supported on the point set P . The Fourier transform of this measure is given by:

$$\hat{\mu}(\xi) = \sum_{p \in P} e^{-2\pi i p \cdot \xi}$$

To count the number of unit distances, we examine the convolution $\mu * \sigma$, where σ is the uniform surface measure on the unit circle $\mathbb{S}^1$. The total number of ordered point pairs at distance exactly 1 can be expressed as:

$$U(P) = \iint_{\mathbb{R}^2 \times \mathbb{R}^2} \sigma(x - y) d\mu(x) d\mu(y)$$

Applying Plancherel's theorem, we transition this integral to the frequency domain:

$$U(P) = \int_{\mathbb{R}^2} |\widehat{\mu}(\xi)|^2 \widehat{\sigma}(\xi) d\xi$$

The Fourier transform of the surface measure σ on $\mathbb{S}^1$ is given by $J_0(2\pi|\xi|)$, where J_0 is the Bessel function of the first kind of order zero. We employ the asymptotic expansion for large argument $|\xi|$:

$$\widehat{\sigma}(\xi) = J_0(2\pi|\xi|) = \sqrt{\frac{2}{\pi(2\pi|\xi|)}} \cos(2\pi|\xi| - \frac{\pi}{4}) + O(|\xi|^{-3/2})$$

4.2 Isolating the Principal Term

We decompose the integral over frequency space into a low-frequency part and a high-frequency part. For a parameter $R > 0$, we write:

$$\int_{\mathbb{R}^2} = \int_{|\xi| \leq R} + \int_{|\xi| > R}$$

On the low-frequency component, the Bessel function is smooth and bounded by 1. Using the Cauchy-Schwarz inequality and Plancherel's theorem again:

$$\begin{aligned} \left| \int_{|\xi| \leq R} |\widehat{\mu}(\xi)|^2 \widehat{\sigma}(\xi) d\xi \right| &\leq \int_{|\xi| \leq R} |\widehat{\mu}(\xi)|^2 d\xi \\ &\leq \left(\int_{\mathbb{R}^2} |\widehat{\mu}(\xi)|^2 d\xi \right)^{1/2} \left(\int_{|\xi| \leq R} |\widehat{\mu}(\xi)|^2 d\xi \right)^{1/2} \end{aligned}$$

By the properties of the empirical measure, the integral over the full domain relates to the total mass, which is n . Thus, the low-frequency contribution is bounded by $n^2 R^{-1}$ under appropriate smoothing.

4.3 Bounding the Oscillatory Integral

For the high-frequency component $|\xi| > R$, we substitute the asymptotic expansion of the Bessel function:

$$\int_{|\xi| > R} |\widehat{\mu}(\xi)|^2 \sqrt{\frac{2}{\pi(2\pi|\xi|)}} \cos\left(2\pi|\xi| - \frac{\pi}{4}\right) d\xi$$

To rigorously bound this, we employ a dyadic decomposition. Let $A_j = \{\xi \in \mathbb{R}^2 : 2^j \leq |\xi| < 2^{j+1}\}$. The integral over A_j is:

$$\int_{A_j} |\widehat{\mu}(\xi)|^2 \frac{\cos(2\pi|\xi| - \pi/4)}{\sqrt{\pi^2|\xi|}} d\xi$$

Using the restriction theorem of Tomas-Stein, the L^2 norm of the Fourier transform of a measure supported on a manifold with non-vanishing curvature is bounded. Specifically, for the circle $\mathbb{S}^1$, we have:

$$\int_{\mathbb{S}^1} |\widehat{f}(\omega)|^2 d\sigma(\omega) \leq C \|f\|_{L^{4/3}(\mathbb{R}^2)}^2$$

Applying this to the empirical measure μ (appropriately mollified to a function f), we can bound the integral over each dyadic annulus. The decay rate of $|\xi|^{-1/2}$ from the Bessel function expansion perfectly offsets the growth in the measure of the annuli, allowing the series to converge when appropriately weighted.

4.4 Combinatorial Incidence Bound

While the Fourier analytic approach provides a spectral perspective, the tightest bounds currently rely on incidence geometry. Let C_p be the unit circle centered at $p \in P$. The total unit distance count equals the number of incidences $I(P, \mathcal{C})$ between the point set P and the family of circles $\mathcal{C} = \{C_p : p \in P\}$. We use the crossing lemma. Construct a graph $G = (V, E)$ where vertices are points in P , and edges are arcs of the circles in $\mathcal{C}$ connecting consecutive points on the same circle. The number of vertices is $|V| = n$. Let $|E| = e$. Since two circles intersect in at most two points, the number of edge crossings $cr(G)$ is at most $2\binom{n}{2} \leq n^2$. By the crossing number inequality, if $e \geq 4n$, then:

$$cr(G) \geq \frac{e^3}{64n^2}$$

Substituting the upper bound for crossings:

$$\frac{e^3}{64n^2} \leq n^2 \implies e^3 \leq 64n^4 \implies e \leq 4n^{4/3}$$

The number of incidences is bounded by the number of edges plus the number of circles, so $I(P, \mathcal{C}) \leq e + n \leq 4n^{4/3} + n$. Thus, the maximum number of unit distances is $O(n^{4/3})$.